

Energy tasks and Brazilian open data

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Tier	#	Task	Stated objective	Open national source	What the source provides
I	2	Generation and load forecasting	Forecast solar generation, wind generation and load. Recent work also uses the task as a test bed for time series foundation models [3, 4]	ONS programacao_x_previsao; carga-energia-programada; curva-carga; demanda_maxima_di	Forecast and scheduled output per plant, in 48 daily steps, for 597 plants. The plant is identified by cod_usinapdp, a third system, with no intersection with id_ons or usina_conjunto; matching by name, with case and whitespace normalized, resolves 26 of the 597
I	10	Asset failure and unavailability	Estimate failure rate, availability and maintenance need of the generating unit [14]	ONS taxa_teif_teip; disponibilidade_usina; ind_disponibilidade_*; ind-disponibilidade-geracao-sin	The TEIFa and TEIP rates cover 265 plants and 251 CEG on a monthly basis, from Jul 2021 to Jul 2026, with version and calculation date. Hourly availability carries installed capacity and operational availability for 158 plants identified by id_ons and ceg
I	12	Interchange, flow and network constraint	Estimate flow between subsystems, transfer limit and energy curtailed by network constraint [17, 24]	ONS intercambio-nacional, intercambio-internacional, intercambio_modalidade, programacao_fluxo_controlado, restricao_coff_eolica_detail and _usi, restricao_coff_fotovoltaiica	National, international and by-modality interchange cover 27 years each, from 2000 to 2026, in one resource per year and format. Controlled flow covers 700 days since 2024, and wind constrained-off, 60 months since Oct 2021, with verified generation against scheduled, plant by plant
I	13	Power quality, frequency and ancillary services	Measure frequency and voltage deviation, and the reserve that keeps them in band in a system of decreasing inertia [18]	ONS ind_qualid_dfd_evento, ind_qualid_dfd_meano, ind_qualid_dfp_regime, ind_confiarb_dreq_freq, uge_operacao_csi, usi_servicoancilar_sandbox, oferta_respostademanda	The four quality and frequency indicators come in a single cumulative file, with no year in the resource name. The synchronous condenser is published since 2021, demand response since 2023, and the reactive ancillary service only in 2026
II	1	Resource assessment	Measure available irradiance and wind, and validate a satellite product against surface measurement [1, 2]	ONS restricao_coff_fotovoltaiica_detail (plane-of-array irradiance); Rede SONDA, INPE; Atlas Brasileiro de Energia Solar, LABREN; NASA POWER	Plane-of-array irradiance has 19,088,880 records from 560 plants in 10 states, every 30 min, from Apr 2024 to Sep 1, 2026, of which 16.9% arrive flagged as suspect. The Atlas carries 72,272 records of monthly and annual means
II	3	Asset inventory and detection	Locate and date plants and distributed installations from satellite imagery [5, 6, 7]	ANEEL SIGA; ANEEL MMGD; ONS linha-transmissao and subestacao; EPE; Sentinel-2	SIGA carries 25,133 projects with coordinate, capacity and commissioning date. MMGD carries 4,678,147 connections, whose coordinate is that of the substation and not of the installation. The network has 2,208 lines and 1,677 substations
II	4	Satellite monitoring of operation	Estimate generation and emissions of a plant without telemetry, calibrating a plume or thermal signature against reported generation [8]	ONS geracao-usina-2; geracao-termica-despacho-2; cvu-usitermica; balanço_dessem_detailhe	Thermal dispatch comes out by reason, on a monthly basis, and unit variable cost per plant, on an annual basis since 2020. The DESSEM balance carries 192 rows per day: four subsystems in 48 steps

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II	9	System adequacy and reliability	Estimate the risk of load not served and the severity of interruption, and measure unserved energy [13]	ONS ind_confiarb_energianaosuprida; interrupcao_carga; energia-vertida-turbinavel; ind_confiarb_* (robustness, severity, CIPER); ear-diario-por-subsistema	Unserved energy has 37,592 records from Jan 2007 to Jul 2026, in 32 aggregation attributes. Load interruption has 8,325 events and 6,821 disturbances from Jan 2007 to Sep 2026, with interrupted load, mean duration and unserved energy
II	11	Price and marginal operating cost	Forecast the half-hourly marginal operating cost and the settlement price, with the metric used in the price forecasting literature [16]	ONS cmo-semanal, cmo-semi-horario, ofertapreco-importacao, cvu-usitermica; CCEE, Dados Abertos (PLD)	The weekly CMO covers 2005 to 2026; the half-hourly, 2020 to 2026, for four subsystems in 48 half-hours per day; and the import price offer, 2019 to 2026. The half-hourly series is dispatch-model output, not ex-post measurement: it is published before the day it refers to
II	15	Methane leakage and fugitive emissions	Detect and quantify a methane point plume from satellite, with detection limit in kg/h as the metric [19]	Sentinel-5P/TROPOMI, Sentinel-2 and EMIT, open; ANP, oil and gas production per well	ANP publishes monthly production per well, in 54 files. It is the only national source that locates the installation as a plume candidate
II	16	Right-of-way vegetation and line risk	Measure vegetation encroachment on the transmission corridor and the associated ignition risk [20]	ONS linha-transmissao; MapBiomass; INPE, Programa Queimadas	INPE publishes active fire detections in 23 annual files, from 2003 to 2025, for the reference satellite, besides the 10 min, daily and monthly series. MapBiomass publishes annual land cover classification for the whole country, in raster. Line geometry comes from ONS, in 2,208 alignments
III	6	Plant performance and degradation	Measure performance ratio, soiling and unavailability over the life of the asset [9, 10]	ONS restricao_coff_fotovoltaica_detail (POA, verified and estimated generation); disponibilidade_usina; taxa_teif_teip	POA, verified generation and estimated generation come out together every 30 min, from Apr 2024 to Sep 1, 2026, which gives the numerator and the denominator of the performance ratio in the same record. Availability and the unavailability rates cover Jul 2021 to Jul 2026
III	7	Siting and land use conflict	Select a site and measure competition with agriculture, including agrivoltaic arrangements [11, 12]	MapBiomass; IBGE; ANEEL SIGA; ONS basin outlines	MapBiomass gives the land use class in a national annual raster; SIGA, the coordinate and capacity of 25,133 projects; and ONS, the basin outline. Crossing site against current use is immediate, and against past use as well, through the annual series
III	8	Energy access and demand	Estimate electrification and demand where there is no metering, from night lights and administrative records [15]	ANEEL MMDG by municipality; ONS curva-carga; VIIRS	MMDG carries the IBGE municipality code in the 4,678,147 connections, which gives a count per municipality without going through coordinates
III	14	Marginal emissions and carbon intensity	Attribute emissions to the marginal MWh, that is, to the generator that responds to a change in load, rather than to the system average [23, 21]	ONS geracao-usina-2, cvu-usitermica, cmo-semi-horario, carga-energia; MCTI/SIRENE, Operating Margin Emission Factor	MCTI publishes the Operating Margin by the Dispatch Data Analysis Method on an hourly basis, with base years from 2006 to 2026. The grid is rectangular: 31 days in every month, and the nonexistent day arrives as zero, not empty. The input is entirely within ONS, and the target outside it

Two priority sources name tasks: the taxonomy of [23], which marks a task as *High Leverage*, and the European Parliament survey on artificial intelligence in the energy sector [22]. **I** task named by both sources; **II** task named by one of the two; **III** task named by neither. The table runs from tier I to tier III. Resource is the CKAN unit: the same period enters once per format, and 27 years of national interchange amount to 60 resources.

Method classes, and where each one applies

- core
- the estimand is defined in the terms of that method: without it, the task does not exist
- auxiliary
- the method enters as input, calibration or constraint
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- empty
- the method does not apply to the task as posed

Tier			Problem structure				Data	Solution class		
			Time series	Tabular panel	Spatial	Graph	Computer vision	Deterministic physics	Machine learning	Optimization
I	2	Forecasting	■	■	□		□	□	■	
I	10	Failure and unavailability	■	■		□	□	□	■	□
I	12	Interchange and flow	■	□	□	■		■	□	■
I	13	Quality and frequency	■	□		□		■	□	□
II	1	Resource assessment	□	□	■			■	□	
II	3	Inventory and detection	■		■	□	■		■	
II	4	Satellite monitoring	□	■	□		■	□	■	□
II	9	Adequacy and reliability	■	□	□	■		■	□	■
II	11	Price and marginal cost	■	□	□	□		□	■	■
II	15	Methane and leakage	□		■		■	■	■	
II	16	Right-of-way and risk	□		■	□	■		■	□
III	6	Performance, degradation	■	■	□			■	□	
III	7	Siting and land use	□		■	□	■	□	□	■
III	8	Access and demand	□	■	■		■		■	
III	14	Marginal emissions	■	■	□	□		□	■	□

The column selection follows [23], with the graph in a column of its own because it constitutes there a method class with its own formulation and metrics [24], and not a particular case of a panel. *Tabular panel* is the structure of the data, plant × time, and *machine learning* is the technique that solves it: a panel can be fitted by a declared regression, and machine learning also operates over imagery and over series.

Nine tasks have a time series core. In five of them, the estimand is already published as a series: performance (6), with POA and generation at 30 min since Apr 2024; adequacy (9) and asset failure (10), whose entire history fits in one file, with unserved energy and load interruption since Jan 2007 and TEIFa and TEIP since Jul 2021; price (11), with the half-hourly CMO since 2020; and marginal emissions (14), with the hourly operating margin since 2006.

In the other four the estimand requires assembly, and the obstacle differs. In forecasting (2) it is linkage, through the identifier proper to the scheduling file. In asset detection (3) it is imagery: the national source gives the inventory, not the series that would date it. In network flow (12) and in frequency quality (13) it is

format, with interchange in one resource per year and the frequency indicators in a single cumulative file, with no year in the name.

None of the fifteen tasks lacks published national data. Task 14 is the only case with the input in ONS and the estimand outside it, and the estimand is published as well: the marginal emission factor comes from MCTI on an hourly basis since 2006. A single input serves three tasks. Plane-of-array irradiance from 560 plants, every half hour, is an input to resource assessment, ground truth for forecasting and an input to the performance ratio. There is no public equivalent in the country: Rede SONDA measures in the horizontal plane, in a network of dozens of stations, and the LABREN Atlas is a modeled product in monthly means.

Open national sources outside ONS

CCEE is independent of the operator, a condition of task 4, which compares a satellite estimate against the reported value: ONS on both sides does not constitute an independent comparison. The remaining sources cover what ONS does not publish: interruption at the distribution level, surface meteorological measurement, active fire detections and land cover.

Agency and dataset	What it provides	Coverage and volume	Tasks	Access route
ANEEL, Dados Abertos: interrupcoes-de-energia-eletrica-nas-redes-de-distribuicao	Every interruption in the country’s distribution network, event by event, outside the areas of permit-holding utilities	One file per year, from 2017 to 2026, in ZIP and Parquet, from 78 MB (2017) to 272 MB (2025)	9	Open CKAN API, no registration
ANEEL, Dados Abertos: indicadores-coletivos-de-continuidade-dec-e-fec	DEC and FEC per set of consumer units, with the compensation paid for violation, the regulatory limit and the physical-electrical attributes of the set	Three ranges, 2000–2009, 2010–2019 and 2020–2029, besides compensation, limit, set attributes and domain table	9	Open CKAN API, no registration
INMET, automatic stations and BDMEP	Hourly surface measurement: temperature, humidity, wind, precipitation and global radiation	674 automatic stations, 518 operating and 156 out of service, distributed over 25 states; RR and SE are absent. The oldest has operated since May 6, 2000	1, 2	Catalog through the API, no registration. The hourly endpoint of apitempo returns 204 No Content and does not serve the series; global radiation comes through the annual ZIP of portal.inmet.gov.br
CCEE, Dados Abertos	Generation and consumption from market settlement, per plant and per source, previously released in the InfoMercado spreadsheets	No measured coverage: the API responds HTTP 403, “Acesso bloqueado”	2, 4, 11, 14	The block applies to all three routes: CKAN direct, CKAN with a Referer from the portal itself, and the host www.ccee.org.br. The portal responds in the browser, and the programmatic route does not
INPE, Programa Queimadas: active fire	Satellite fire detections, event by event with date, time and coordinate	Annual ZIP file from 2003 to 2025 in four extents: Brasil-sat-ref, Brasil-todos-sats, AMS-sat-ref and EstadosBr-sat-ref	16	Open file server, no registration. The path anual/Brasil/ returns 404; the four extents above are the ones that exist
MapBiomass, land cover and use collection	Annual land cover and use classification for the whole country, in raster	Collection 11 at 30 m occupies 763 MB per year, and collection 4 at 10 m, 4.7 GB, in BigTIFF	7, 16	GeoTIFF direct from the public bucket, no registration and without going through Earth Engine

File size is the one declared by the resource, except for INPE and MapBiomass, measured in the HTTP header.

The target of task 14 is at MCTI. Task 9 has two scales of interruption: the transmission scale, from ONS, and the distribution scale, event by event since 2017 and as a set indicator since 2000. INMET gives task 1 a further 518 operating surface stations, against the dozens of Rede SONDA; they measure in the horizontal

plane and not in the plane of the array, and therefore complement the ONS POA without replacing it. Task 16 goes from the isolated line geometry to the crossing with the corridor, through INPE fire detections and MapBiomass land cover, both without registration; MapBiomass also serves task 7.

The data cell gives the coverage the source declares — period, granularity, entity count —, and this establishes neither a continuous series nor a filled column. The column name can also mislead: in val_irradianciaverificado, it diverges from the quantity the field carries.

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